Resonant cascaded Kerr nonlinearity in $\chi^{(2)}$ metasurfaces

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Second-harmonic generation followed by difference-frequency generation produces an effective third-order response at the pump frequency, and in bulk crystals its strength is set by the phase mismatch. We show that a resonance at the second harmonic, with quality factor Q_2 and detuned by $\Delta\omega$ from twice the pump frequency, takes the place of the phase mismatch and enhances the cascaded susceptibility in proportion to Q_2. We derive a closed expression for the cascaded susceptibility of a resonator from the Green's function of the second-harmonic field and from coupled-mode theory; for a uniform medium it reduces to the bulk result. The cascaded response changes sign with $\Delta\omega$, is accompanied by a nonlinear loss, and obeys a bandwidth sum rule that limits it to narrowband operation. In a doubly resonant $LiNbO_3$ metasurface with $Q_2 = 4000$ and an overlap factor of 0.3, the cascaded susceptibility exceeds the native Kerr susceptibility of the host by a factor of about 50. We propose to detect the sign change with a picosecond Z-scan as a function of pump wavelength in a singly resonant metasurface. In a periodically poled thin-film $LiNbO_3$ microring the same theory gives a factor of about 10^3 and microwatt thresholds for bistability and parametric oscillation.

I. INTRODUCTION

1. Kerr nonlinearity underlies all-optical switching, optical bistability, parametric oscillation and frequency combs. The native $\chi^{(3)}$ of low-loss dielectrics is small, about 2×10^{-21} m^2/V^2 in $LiNbO_3$, and field enhancement in resonant nanostructures increases the nonlinear phase per unit power without changing the material susceptibility.

2. Two cascaded second-order processes act on the pump as a Kerr medium. The effect was predicted by Ostrovskii [1], observed in KTP as a nonlinear refraction that changes sign with the phase mismatch [2], reviewed in Ref. [3], and described microscopically as a collective effect [4]. In bulk crystals its strength scales as $1/\Delta k$, and it is used for soliton compression in $LiNbO_3$ [5–7].

3. Metasurfaces and nanoresonators with bound states in the continuum enhance second-harmonic generation [8–12], and cascaded second-order processes in nanostructures have produced third-harmonic light [13, 14]. In metasurface second-harmonic generation the back-conversion of the second harmonic has been treated as pump depletion [15]. In $\chi^{(2)}$ microresonators the cascaded Kerr response has been tuned through the detuning of the second-harmonic mode [16–18]. We are not aware of a treatment of the metasurface cascade as a Kerr response on the pump that compares it with the native response of the host.

4. Here a detuned second-harmonic resonance replaces the phase mismatch, so that the cascaded susceptibility scales as $Q_2/\Delta\omega$, changes sign with $\Delta\omega$ and comes with a nonlinear loss. It reaches about fifty times the native Kerr susceptibility of $LiNbO_3$ in a doubly resonant metasurface and about 10^3 in a periodically poled microring.

II. CASCADED SUSCEPTIBILITY OF A RESONATOR

A. Fields, susceptibilities and modes

Real fields are written as $\mathbf{E}e^{-i\omega t}$ + c.c., so that the second-harmonic and difference-frequency polarisations are $\mathbf{P}(2\omega_1) = \varepsilon_0\chi^{(2)}\mathbf{E}_1\mathbf{E}_1$ and $\mathbf{P}(\omega_1) = 2\varepsilon_0\chi^{(2)}\mathbf{E}_2\mathbf{E}_1^*$, and self-phase modulation is $3\varepsilon_0\chi^{(3)}|\mathbf{E}_1|^2\mathbf{E}_1$ [19]. The tensor $\chi^{(2)}_{ijk}$ obeys Kleinman symmetry. The modes $\mathbf{f}_n$ of the linear structure at ω_1 and $\omega_2 \simeq 2\omega_1$ are normalised to unit stored energy with the dispersive energy density, which contains $\tilde{\varepsilon}_n = \partial(\omega\varepsilon_r)/\partial\omega$ at ω_n, and $N_n = \frac{1}{2}\int(\varepsilon_0\tilde{\varepsilon}_n|\mathbf{f}_n|^2+\mu_0|\mathbf{h}_n|^2)\,d^3r$ is the norm that appears when Maxwell's equations are projected onto a mode.

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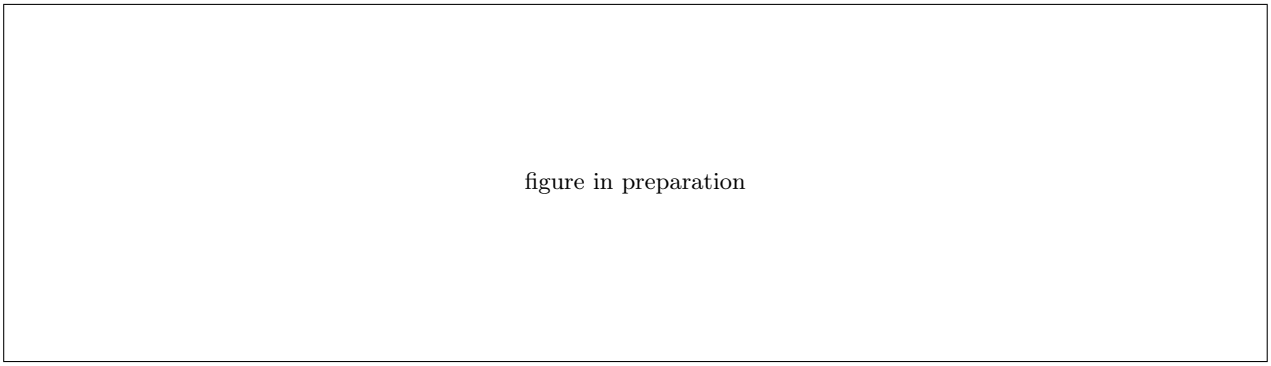


FIG. 1. Cascaded Kerr response at a second-harmonic resonance. (a) Second-harmonic generation and difference-frequency generation through a mode at ω_2, detuned by $\Delta\omega = 2\omega_1 - \omega_2$. (b) Doubly resonant $LiNbO_3$ metasurface with a quasi-bound state at ω_1 and a symmetry-protected mode near $2\omega_1$ in one meta-atom. (c) Periodically poled microring with pump and second-harmonic modes. (d) Real (solid) and imaginary (dashed) parts of $\chi^{(3)}_{\mathrm{casc}}$ versus $\Delta\omega/\gamma_2$, normalised to the value at $\Delta\omega = 2\gamma_2$.

B. Green's-function derivation

The second-harmonic field driven by $\mathbf{P}(2\omega_1)$ is written with the dyadic Green's function of the structure, which gives a cascaded polarisation $\chi^{(2)} \cdot G(2\omega_1) \cdot \chi^{(2)}\, \mathbf{E}_1\mathbf{E}_1\mathbf{E}_1^*$ with a nonlocal kernel. Near an isolated second-harmonic quasinormal mode the pole term of G dominates [20, 21], and the kernel separates into two overlap integrals of the form used for second-harmonic generation in nanoresonators [22].

C. Coupled-mode derivation

Projecting Maxwell's equations onto the two modes [23, 24] gives equations for the mode amplitudes a_1 and a_2, with the second-harmonic and difference-frequency couplings $\beta = \varepsilon_0\omega_1 M/N_2$, equal for $N_1 = N_2$ as required by the Manley–Rowe relations, and the native coupling α_3. In steady state the second-harmonic amplitude follows the pump algebraically, so the cascaded coefficient $-|\beta|^2/(\Delta\omega + i\gamma_2/2)$ holds at any pump amplitude below depletion. For a time-dependent pump the Kerr description requires a pump bandwidth small compared with $|\Delta\omega + i\gamma_2/2|$.

D. Effective susceptibility

Both derivations give

$$\chi^{(3)}_{\mathrm{casc}} = -\frac{2\varepsilon_0\omega_1}{3N_2}\,\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int |\mathbf{f}_1|^4\, d^3r} \simeq -\frac{2\omega_1}{3n_2^2\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}, \tag{1}$$

with $M = \int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}\, d^3r$, γ_2 the energy decay rate and n_2 the effective index of the second-harmonic mode, and $\eta_{\mathrm{geom}} = \varepsilon_0 n_2^2 |M|^2/(|\chi^{(2)}|^2 N_2 \int |\mathbf{f}_1|^4\, d^3r) \le 1$ a geometric overlap. The powers of n_1 cancel between $|M|^2$ and $\int |\mathbf{f}_1|^4$, and the result does not depend on how the modes are normalised. For phase-matched plane waves in a uniform medium, $\eta_{\mathrm{geom}} = 1$ and Eq. (1) becomes the bulk bound-wave susceptibility $-2|\chi^{(2)}|^2/[3(n_2^2 - n_1^2)]$; the frequency mismatch $\Delta\omega$ then plays the role of the wavevector mismatch of bulk cascading.

E. Range of validity

The derivation assumes one second-harmonic mode, $|\Delta\omega| \gtrsim \gamma_2$ and an undepleted pump. Cascaded optical rectification and the linear electro-optic effect [25], and the longitudinal part of the bound second-harmonic field, contribute terms that are not resonantly enhanced and add a constant background. Related effective third-order responses have been derived for one-dimensional photonic crystals [26], for local-field corrections [27] and for broadband quantum optics [28].

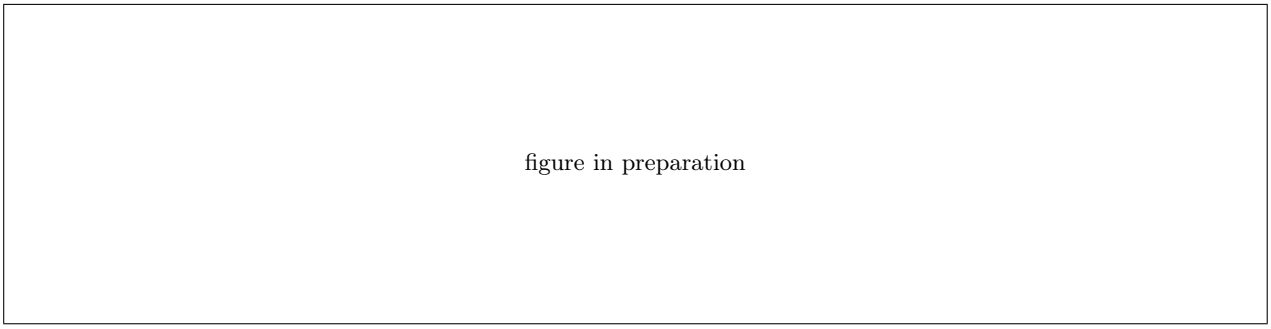


FIG. 2. Checks of Eq. (1). (a) Kerr coefficient of the pump mode from integration of the two-mode equations (markers) and from the closed form (lines) versus $\Delta\omega/\gamma_2$. (b) Second-harmonic amplitude from the full driven field of a layered Bloch cell divided by the single-mode prediction, versus $\Delta\omega/2\omega_1$. (c) Peak $|\chi^{(3)}_{\mathrm{casc}}|$ (left axis) and its bandwidth (right axis) versus Q_2 at fixed $\Delta\omega/\gamma_2$.

TABLE I. Cascaded enhancement at $Q_2 = 4000$, $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$, at telecom wavelengths.

Host	$\lvert\chi^{(2)}\rvert$ (pm/V)	$\chi^{(3)}_{\mathrm{nat}}$ (m^2/V^2)	n_2	R	Q_2^{cross}
$LiNbO_3$	50	2×10^{-21}	2.26	49	82
GaP	100	1×10^{-19}	3.1	2.1	1.9×10^3
AlN	10	2×10^{-21}	2.1	2.3	1.8×10^3
GaAs	350	1.4×10^{-18}	3.5	1.4	2.8×10^3
AlGaAs	100	1×10^{-18}	3.3	0.18	2.2×10^4

III. LIMITS OF THE RESONANT CASCADE

The imaginary part of $\chi^{(3)}_{\mathrm{casc}}$ is even in $\Delta\omega$ and the real part is odd. Their ratio $\rho = \gamma_2/(2|\Delta\omega|)$ equals 1/4 at $\Delta\omega = 2\gamma_2$, and bistability and parametric gain both require $\rho < 1/\sqrt{3}$.

The peak susceptibility grows as Q_2 and its bandwidth falls as γ_2, so that $|\chi^{(3)}_{\mathrm{casc}}|_{\max}\,\Delta f \lesssim \omega_1 |\chi^{(2)}|^2 \eta_{\mathrm{geom}}/(3\pi n_2^2)$ independently of Q_2. The resonant cascade is therefore suited to narrowband and continuous-wave operation. Trade-offs between strength and bandwidth of resonant nonlinear enhancement are discussed for plasmonic and native nonlinearities in Ref. [29], for harmonic generation in Ref. [30], and for a cavity-enhanced cascaded frequency shifter in Ref. [31].

Pump depletion sets in when $|\beta a_1|$ approaches $|\Delta\omega|$, above which the steady state becomes parametrically unstable [32]; at fixed $\Delta\omega/\gamma_2$ the largest stored pump energy scales as Q_2^{-2}.

Phase-mismatched bulk PPLN gives $|\chi^{(3)}_{\mathrm{casc}}| \approx 10^{-20}$ m^2/V^2 over about 10 THz, and a doubly resonant metasurface with $Q_2 = 4000$ gives about 10^{-19} m^2/V^2 over 100 GHz. Bulk cascading is ahead in $|\chi^{(3)}|\,\Delta f$ by about an order of magnitude, while the metasurface uses a sub-micron layer and reaches its operating point with continuous-wave pumping. Native Kerr switching in quasi-bound-state metasurfaces reaches femtosecond speed [33], far below the picosecond response time $1/\gamma_2$ of the resonant cascade.

IV. CASCADED AND NATIVE KERR RESPONSE

The enhancement ratio $R = |\chi^{(3)}_{\mathrm{casc}}|/|\chi^{(3)}_{\mathrm{nat}}|$ depends on the host through $|\chi^{(2)}|^2/(n_2^2\chi^{(3)}_{\mathrm{nat}})$ and on the resonator through $Q_2\eta_{\mathrm{geom}}$ and $\Delta\omega/\gamma_2$. At $\Delta\omega = 2\gamma_2$, $R = Q_2|\chi^{(2)}|^2\eta_{\mathrm{geom}}/(6n_2^2\chi^{(3)}_{\mathrm{nat}})$, and the cascade dominates above $Q_2^{\mathrm{cross}} = 6n_2^2\chi^{(3)}_{\mathrm{nat}}/(|\chi^{(2)}|^2\eta_{\mathrm{geom}})$. $LiNbO_3$ combines a moderate $\chi^{(2)}$ with a weak native Kerr response (Table I). GaP, AlN and GaAs are close to break-even at $Q_2 = 4000$, and the band-edge enhancement of the native response makes AlGaAs unfavourable at telecom wavelengths. With a common pump resonance the same R multiplies the per-photon nonlinear phase, so the cascade adds a factor Q_2 to the Q_1^2 scaling of the native architecture.

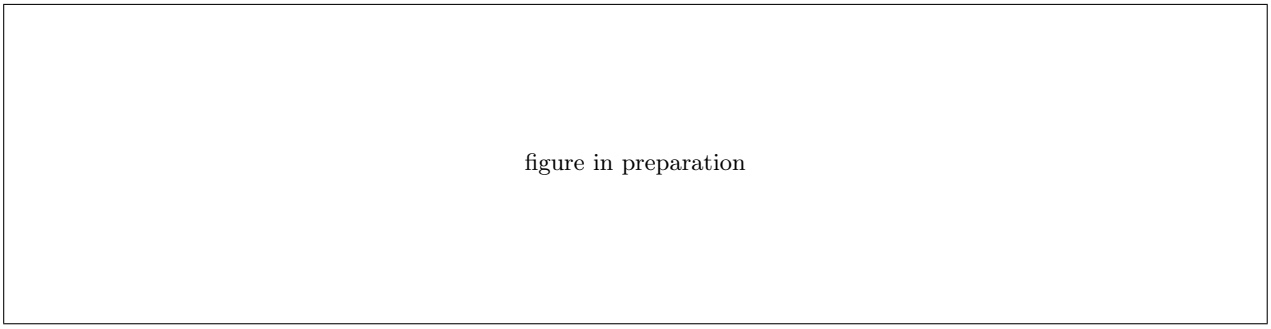


FIG. 3. (a) R versus Q_2 at $\Delta\omega = 2\gamma_2$ and $\eta_{\mathrm{geom}} = 0.3$ for the hosts of Table I; triangles mark Q_2^{cross}. (b) R for $LiNbO_3$ versus Q_2 and η_{geom}, with the doubly resonant design point, the singly resonant design of Sec. V C and the poled microring of Sec. VI marked.

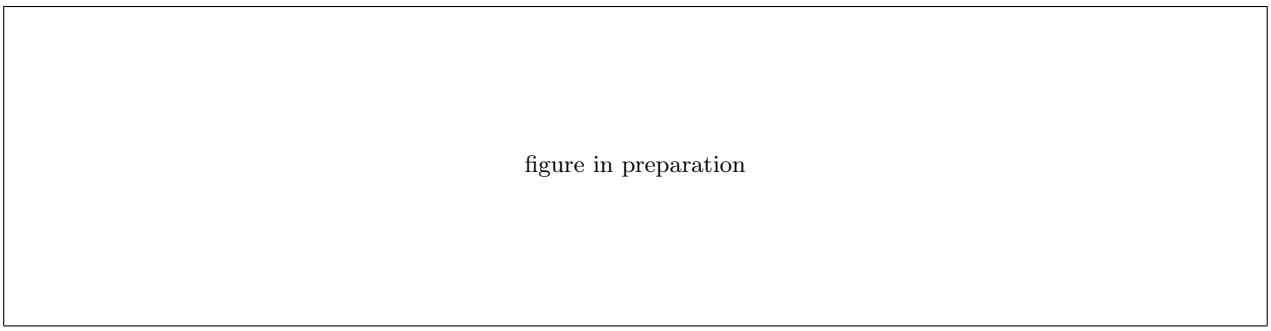


FIG. 4. Two-mode coupled-mode simulations of a doubly resonant metasurface with $\Delta\omega = -2\gamma_2$, $\gamma_1 = 10\gamma_2$ and $R = 49$. (a) Stored pump energy versus input power for increasing and decreasing sweeps. (b) Cascaded phase and onset of parametric oscillation versus input power. (c) Ratio of four-wave-mixing efficiencies with and without the second-harmonic mode, with $(1 + R)^2$ as a dashed line.

V. METASURFACE IMPLEMENTATION

A. Dark second-harmonic resonance

A symmetry-protected bound state in the continuum near $2\omega_1$ has a quality factor limited by absorption and fabrication [9], and a quasi-bound state at ω_1 couples the pump from free space [8, 10]. Doubly resonant structures with a bound state at the second harmonic have been designed and measured for second-harmonic generation [34, 35]. The overlap M is nonzero only if the unit-cell symmetry of $\mathbf{f}_2$ matches that of the $\chi^{(2)}$ tensor contracted with $\mathbf{f}_1\mathbf{f}_1$; we give the selection rules for the orientations of the d_{33} axis in a $LiNbO_3$ film [12]. At normal incidence both modes lie at the Γ point, which removes the bulk phase-matching condition. The detuning $\Delta\omega$ is set by the geometry and tuned by temperature or by the electro-optic effect. Fabrication disorder and finite array size lower the collective quality factor of the second-harmonic mode, and we estimate both from supercell simulations.

B. Doubly resonant metasurface

With a common pump resonance the cascaded and native architectures share the field enhancement, and each observable scales with a fixed power of $1 + R$: self- and cross-phase modulation as $1 + R$, the bistability threshold and switching energy as $1/(1 + R)$, and the four-wave-mixing efficiency as $(1 + R)^2$. Cascaded bistability has been observed in a monolithic $LiNbO_3$ resonator with a phase-mismatched second harmonic [36]. For a $LiNbO_3$ design with $Q_2 = 4000$, $\eta_{\mathrm{geom}} = 0.3$ and $\Delta\omega = 2\gamma_2$, $R \approx 49$. The cascade also gives an intensity-dependent absorption tuned by $\Delta\omega$, usable as an optical limiter, and a transmission response to modulation of $\Delta\omega$.

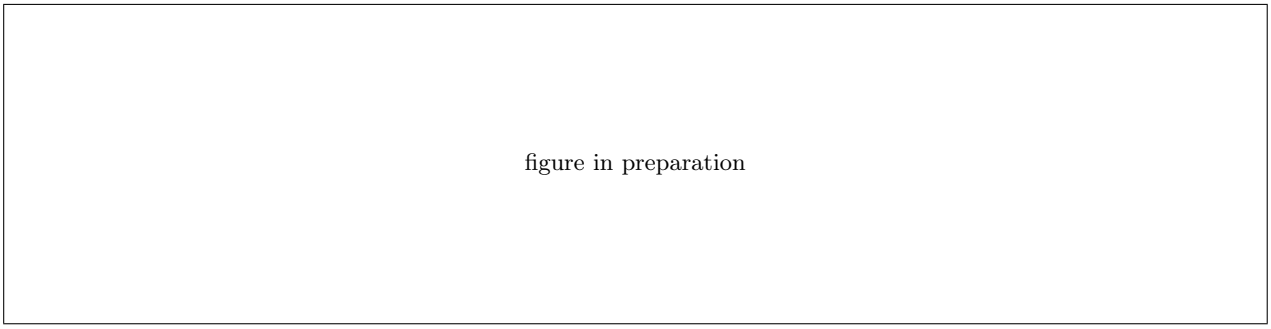


FIG. 5. Singly resonant $LiNbO_3$ metasurface. (a) Unit cell and second-harmonic mode profile. (b) Transmission at ω_1 and $2\omega_1$, and $|\widetilde{M}|$, versus frequency. (c) Real and imaginary parts of $\chi^{(3)}_{\mathrm{casc}}$ versus $\Delta\omega/\gamma_2$. (d) Z-scan phase for picosecond pulses, relative to its continuous-wave value, versus pulse duration.

C. Singly resonant metasurface

Without a pump resonance the pump field in the meta-atom is fixed by the incident wave. The overlap becomes $\widetilde{M} = \int \chi^{(2)}_{ijk} f^*_{2,i} e_{1,j} e_{1,k}\, d^3r$, with $\mathbf{e}_1$ the linear pump field per unit incident amplitude, and the unit-cell volume replaces $\int |\mathbf{f}_1|^4$. The susceptibility keeps the form of Eq. (1). The per-photon phase loses the Q_1^2 factor of the pump resonance, and bistability and switching are absent. We evaluate one fabricable $LiNbO_3$ design with full-wave simulations, compute $\widetilde{M}$ from its fields, and give $\chi^{(3)}_{\mathrm{casc}}(\Delta\omega)$, R and the single-pass phase; designs simulated so far reach R between 2 and 10. In pulsed operation the pulse must be longer than the second-harmonic lifetime $1/\gamma_2$, and its bandwidth must stay below about $2\gamma_2$ for the generated second harmonic to stay on one side of the resonance. Time-domain coupled-mode simulations give the reduction of the Z-scan phase with pulse duration.

D. Proposed measurement

The sign change of the cascaded nonlinear refraction would be measured with a closed-aperture Z-scan as a function of pump wavelength, which moves $2\omega_1$ across ω_2, in analogy with the angle-tuned measurement in KTP [2]. Near a Mie resonance the native response alone can change the sign of the effective nonlinear index [37]. The cascaded part is the component odd in $\Delta\omega$ at fixed pump detuning, which is well defined when the pump transmission is flat across the scan. The second-harmonic conversion efficiency of the same sample would fix the absolute value of $\widetilde{M}$.

VI. MICRORING RESONATORS

The same elimination applies to a doubly resonant $\chi^{(2)}$ microring, where it reproduces the cavity expression used in Refs. [16, 17], and the measured second-harmonic efficiency fixes the coupling rate g [38–40]. For periodically poled thin-film $LiNbO_3$ with a loaded Q_2 of 3–5 $\times 10^5$, $R \approx 2.8 \times 10^3$ for the device of Ref. [39] and 1.8×10^3 for a smaller ring, against about 13 without poling. Since $Q_2 < 2Q_1$, the second-harmonic linewidth exceeds the pump linewidth, and the cascade lowers the modulation bandwidth by 2 to 7 percent. Bistability and parametric oscillation move from about 14 mW and 9 mW for the native response of the same ring to about 35 μW and 8 μW. The bistability threshold is evaluated at the pump-frame mismatch $\Delta\omega - 2\delta_1$, with δ_1 the pump detuning, and the parametric threshold follows from a multimode stability analysis; parametric oscillation thresholds of tens of microwatts have been measured in thin-film $LiNbO_3$ [41, 42], and a four-wave mixing in a poled waveguide corresponds to an effective Kerr index 1.6×10^4 times the intrinsic one [43]. Thermal and photorefractive shifts [44] are comparable to the cascaded shift at these powers. To leading order they are even in $\Delta\omega$, and an electro-optic dither of $\Delta\omega$ with lock-in detection at $\delta_1 = 0$ would separate them from the odd cascaded shift. The ring supports continuous-wave switching at femtojoule intracavity energy, cancellation of its native Kerr response, and a choice of the sign of the Kerr term for modulation instability and combs [45, 46]. The dispersive anharmonicity scales as $(g/\kappa_1)(g/\kappa_2)$, with κ_n the half-width of mode n, which is smaller than the resonant figure of merit g/κ_1 used for single-photon nonlinear optics [39].

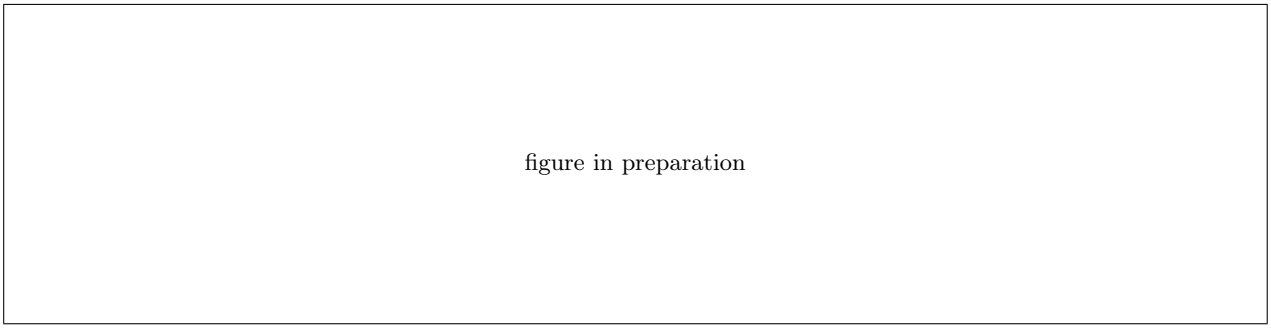


FIG. 6. Periodically poled $LiNbO_3$ microring. (a) R versus Q_2 at $\Delta\omega = 2\gamma_2$ for poled (solid) and unpoled (dashed) rings; the star marks the device of Ref. [39], and horizontal lines mark the doubly resonant metasurface and the singly resonant design. (b) Cascaded and native Kerr shift per photon versus $\Delta\omega/\gamma_2$. (c) Transmission at 30 μW input for $\Delta\omega = \pm 2\gamma_2$ and for the native response. (d) Bistability and parametric thresholds versus Q_2, cascaded and native.

TABLE II. Cascaded Kerr response on four platforms. R is quoted at $\Delta\omega = 2\gamma_2$.

Platform	Cascade set by	R	Limiting bandwidth	Continuous-wave device
Bulk PPLN, phase mismatched	Δk	≈ 5	≈ 10 THz	no
Doubly resonant metasurface	$\Delta\omega$, $Q_2 \approx 4 \times 10^3$	≈ 50	≈ 100 GHz	marginal
Singly resonant metasurface	$\Delta\omega$, $Q_2 \approx 10^3$–6×10^3	2–10	0.1–0.4 THz	no
Poled $LiNbO_3$ microring	$\Delta\omega$, $Q_2 \approx 3 \times 10^5$	$\approx 2 \times 10^3$	≈ 0.2 GHz	yes

VII. DISCUSSION

The platforms in Table II differ in what sets the cascade and in the bandwidth that remains. Engineered second-order media raise R through $|\chi^{(2)}|^2$: coupled quantum wells reach an interband $\chi^{(2)}$ above 10^3 pm/V [47], and quantum-well metasurfaces give a large free-space-accessible second-order response [48]. The gain holds if the native third-order response, which is resonantly enhanced in intersubband systems [49], stays small, which requires suppressing the direct third-order pathways in the level design.

VIII. CONCLUSIONS

A detuned second-harmonic resonance turns the cascaded second-order response into a Kerr nonlinearity that changes sign with $\Delta\omega$ and, for $LiNbO_3$, exceeds the native response by about 50 in a doubly resonant metasurface and about 10^3 in a poled microring, within a bandwidth set by the second-harmonic linewidth. Outlook items: the doubly resonant metasurface as a continuous-wave Kerr element, sign control of the Kerr term in microresonator combs, and engineered second-order media in which the native third-order response is suppressed.

DATA AVAILABILITY

The scripts that produce the figures and tables will be made available with the paper.

Appendix A: Modal projection and normalisation

Projection of Maxwell's equations with dispersion onto a resonant mode, the norm N_n, the coupling constants β and α_3, and the Manley–Rowe relations.

Appendix B: Quasinormal-mode Green's function and the cascaded kernel

Pole expansion of $G(2\omega_1)$ for a leaky second-harmonic mode, the nonlocal kernel $\chi^{(2)} \cdot G \cdot \chi^{(2)}$, and its reduction to Eq. (1).

Appendix C: Uniform-medium limit and a layered-cell test

Bound second-harmonic wave in a uniform medium, its longitudinal part, and a comparison of Eq. (1) with the full driven field of a one-dimensional layered cell (Fig. 2b).

Appendix D: Coupled-mode simulations

Equations, integrator, tolerances, run lengths and the convergence criterion for Figs. 2 and 4, and the time-domain model for pulsed excitation in Fig. 5d.

Appendix E: Singly resonant metasurface

Definition of $\widetilde{M}$, its extraction from full-wave fields, the unit-cell average, and the parameters of the design in Fig. 5.

Appendix F: Microring

Photon-number form of the coupled-mode equations and the relation between g and the normalised second-harmonic efficiency, the multimode stability analysis for the parametric threshold, the thermal and photorefractive model, and the electro-optic tuning of $\Delta\omega$.

Appendix G: Parameters

Material constants, mode parameters and design points used in all figures and tables.

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